$CH_3COO^- \} H^+$ $\quad$ $H_2O + \frac{1}{2}O_2 + 2e^-$

Carboxylic acid
CH_3-COOH^+
(organic)

Corrosive environment

1 Air

Hot > Cold

Polluted > pure

Moist humid > Dry

2 water

Hot > Cold

Polluted > pure

salt > soft fresh (Contain low salts) drinking

3 Acids (H^+)

Mineral — very strong — HCl, H_2SO_4, HNO_3, H_3PO_4

>>

organic — weak — formic acid ~~H-COOH~~ ~~CH_3COOH~~ Acetic acid

>>

fatty — very weak — oil, animals or Plants

N.B Mineral Acids is the Most Corrosive

4 Bases (OH^-)

Strong >> weak

KOH, $NaOH$, $Ca(OH)_2$, $Al(OH)_3$

$NH_4OH \rightleftharpoons NH_4^+ + OH^-$
Ammonia

(Note) except steel + strong base (Passivators) → Passive

5 gases

Acidic >> Basic

SO_x, NO_x, H_2S

$NH_{3(g)}$

6 Soil

water, Acids, gases, Pollutant, Bacteria, Minerals, Stray Current